

# THILAKAR RAJ S

TECHNICAL LEAD ENGINEER | SOFTWARE ARCHITECTURE | JAVA & SPRING BOOT | MICROSERVICES

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## PROFESSIONAL SUMMARY

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Technical Lead Engineer with 11+ years of software engineering experience across healthcare, medical diagnostics, logistics, enterprise SaaS, mobile applications, and AI-driven automation. Hands-on in Java, Spring Boot, microservices, REST APIs, React, React Native, AWS, Docker, MySQL, PostgreSQL, MongoDB, CI/CD, automated testing, and application security. Experienced in technical ownership, solution design, system integration, performance engineering, code reviews, release governance, production support, stakeholder communication, and mentoring engineers.

## CORE TECHNICAL SKILLS

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**Technical Leadership:** Technical Ownership, Solution Design, Architecture Reviews, Code Reviews, Technical Governance, Mentoring, Requirement Analysis, Risk Management, Production Support

**Backend:** Java 8-17, Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs, Microservices, Asynchronous Processing

**Frontend & Mobile:** React, JavaScript, TypeScript, React Native, Bluetooth Low Energy (BLE), Offline-First Architecture, SQLite, Thymeleaf

**Databases & Data:** MySQL, PostgreSQL, MongoDB, Redis, SQLite, ETL, Query Optimization, Indexing, Caching, Batch Processing

**Cloud & DevOps:** AWS, Docker, Linux, Nginx, Tomcat, Git, GitHub Actions, CI/CD, Release Automation

**Quality & Security:** Playwright, TypeScript Automation, Page Object Model, SAST, Semgrep, SonarQube, Dependency Scanning, Secret Scanning, Security Testing

**AI & LLM:** OpenAI, Claude, Gemini, RAG, Prompt Engineering, Model Context Protocol (MCP), AI Agents, LLM Orchestration

## PROFESSIONAL EXPERIENCE

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**Voyage Software Technologies Pvt Ltd | Chennai, India**

**Technical Lead Engineer | Jan 2025 – Present**

Lead technical delivery across healthcare, diagnostics, logistics, SaaS, and AI initiatives, covering solution design, implementation, code quality, release readiness, and production support.

Work with developers, QA, business analysts, DevOps, and product stakeholders to break down requirements, identify technical risks, review designs, and drive delivery through the SDLC.

Define coding, branching, code review, CI/CD, testing, and release practices that support consistent engineering quality across projects.

Guide engineers on architecture, debugging, automation, performance, secure development, and maintainable implementation.

**SNAP Healthcare — Revenue Cycle Management Platform | Enterprise SaaS | Healthcare**

- Provided technical leadership for an enterprise healthcare RCM platform, covering planning, technical design, implementation, quality assurance, release management, and production support.
- Led development of a Playwright and TypeScript automation framework for UI and API testing using Page Object Model, reusable components, reporting, and CI/CD integration.
- Established GitHub branching, pull request, code review, merge validation, quality gate, and deployment practices across delivery teams.
- Integrated security checks using Semgrep, SAST, dependency scanning, and secret scanning, with penetration-test remediation tracking.
- Mentored engineers on automation, Agile delivery, debugging, code quality, and release readiness.

**Allergy Vitae — Bioinformatics & Allergy Diagnostics Platform | Java, Spring Boot 3.x, MySQL 8, Async Processing**

- Led design and development of a clinical allergy diagnostics platform covering ETL, bioinformatics calculations, antigen mapping, patient processing, and PDF report generation.
- Designed an ETL workflow combining patient configuration, raw instrument data, and demographic files into a validated processing pipeline with error reporting.
- Designed calculation engines for IgE, IgG, IgG4, C3b, and IgA assays, including percentile analysis, grading rules, IQR, reference upper limits, and z-score thresholds based on scientific requirements.
- Engineered high-volume processing using asynchronous execution, thread pools, batch operations, and SQL optimization for large laboratory datasets.

- Improved MySQL performance through indexing, query tuning, and bulk database operations while working with bioinformatics specialists.

#### **Just Integrate IT Pvt Ltd | Chennai, India**

##### **Senior Software Engineer | Oct 2021 – Dec 2024**

Worked on the design, development, and modernization of enterprise logistics applications using Java, Laravel, Spring Boot, Microservices, MySQL, MongoDB, Redis, and REST APIs.

Played a key role in developing business-critical logistics modules, designing scalable REST APIs, modernizing legacy applications, optimizing backend performance, and supporting production deployments.

Conducted code reviews, mentored junior developers, collaborated with product owners and business stakeholders, and followed Agile development practices.

ShipCRM / Soluship — Multi-Carrier Shipping & Logistics Platform | Java, Spring, Web Services, MySQL, MongoDB, Solr, Linux

- Designed and modernized a multi-tenant shipping platform integrating carriers including UPS, DHL, FedEx, and Purolator, with role-based administration and rate-driven shipment and invoice workflows.
- Integrated Solr search and Apereo CAS single sign-on; improved application performance through query tuning and Nginx load balancing on Linux servers.
- Developed a Java ODBC connector bridging SQL Server, Oracle, and IBM DB2 data sources with the platform through SOAP and REST services.

#### **Just Integrate IT Pvt Ltd | Chennai, India**

##### **Software Engineer | Nov 2014 – Dec 2021**

- Built web services and e-commerce integrations for Shopify, WooCommerce, and Magento.

Contributed to enterprise logistics applications using Java, Struts2 Framework, REST APIs, MySQL, and MongoDB.

Developed and maintained business modules for shipment processing, quotation management, invoicing, shipment tracking, pickup requests, EDI-based bulk shipment processing, customer management, and related enterprise workflows.

Participated in application modernization, third-party integrations, production support, bug fixing, performance optimization, root cause analysis, and Agile delivery.

#### **ARCHITECTURE & ENGINEERING HIGHLIGHTS**

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- Microservices and scalable REST API design
- ETL pipelines and high-throughput asynchronous processing
- Healthcare SaaS and medical-device integrations
- Offline-first mobile architecture and BLE communication
- Database indexing, query optimization, and caching
- PDF generation and dynamic reporting
- Bioinformatics calculation engines
- AI/LLM orchestration, RAG, MCP, and AI agents

#### **LEADERSHIP & DELIVERY**

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Technical ownership | Architecture reviews | Requirement analysis | Code reviews | Sprint and release planning | Stakeholder communication | Technical mentoring | Risk management | Production support | Quality engineering | Security governance | CI/CD and release governance

#### **EDUCATION**

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Master of Computer Application (MCA) — St. Joseph's College, Trichy | 2011–2014

B.Sc. Computer Science — C.S.I. Jayaraj Annapackiam College, Nallur, Manonmaniam Sundaranar University | 2007–2010